Section 2 layout check

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ABSTRACT

1. Study area and data

Placeholder text for column width. The WBEA network (Fig. 1) comprises twelve stations considered in this study (Table 1); datasets and temporal design are summarised in Table 2, and data availability in Fig. 2.

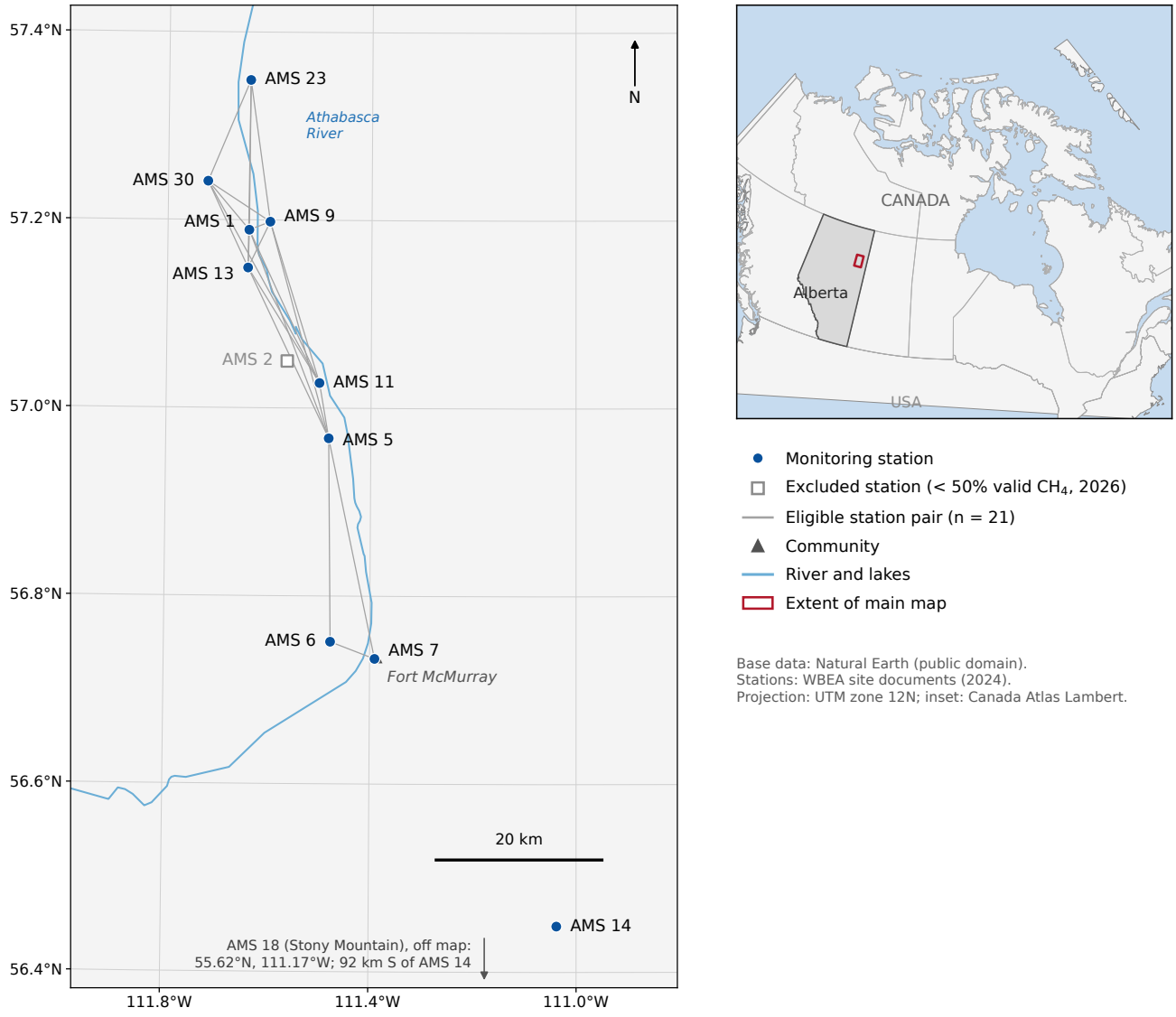


Figure 1: Geographical distribution of the Wood Buffalo Environmental Association atmospheric methane monitoring stations in northern Alberta, Canada. Station locations are shown using official monitoring-network coordinates. Lines connecting monitoring stations represent eligible statistical comparisons rather than verified methane transport pathways.

Table 1
Geographic characteristics and meteorological measurement heights of the WBEA atmospheric monitoring stations included in the study.

Station	Name	Latitude (°N)	Longitude (°W)	Wind height (m)	Temperature height (m)
AMS 1	Fort McKay - Bertha Ganter	57.1894	111.6406	10	2, 10
AMS 2	Mildred Lake	57.0500	111.5641	10	2
AMS 5	Mannix	56.9680	111.4821	20, 45, 75, 90	2, 20, 45, 75, 90
AMS 6	Patricia McInnes	56.7514	111.4767	10	2
AMS 7	Athabasca Valley	56.7334	111.3905	10	2
AMS 9	Barge Landing	57.1982	111.5995	10, 20	2
AMS 11	Lower Camp	57.0268	111.5008	10	2
AMS 13	Fort McKay South	57.1492	111.6423	10	2
AMS 14	Anzac	56.4489	111.0380	20	2
AMS 18	Stony Mountain	55.6215	111.1728	10, 20	2
AMS 23	Fort Hills	57.3489	111.6397	10	2
AMS 30	Ells River	57.2412	111.7221	10	2

Coordinates (WGS84) from WBEA 2024 site documents. Heights (m above ground) list variables with at least 5% valid hourly values in the 2021–2025 exports; where 10 and 20 m are both listed, the 10 m sensor is used when valid. Methane, total hydrocarbons and non-methane hydrocarbons are reported together at every station (ppm, hourly) by WBEA's continuous hydrocarbon analysers, with an automated zero/span check that occupies one hour per day and is flagged in the data (WBEA SOP-ANA-008; WBEA monthly reports). The analyser model is documented here only for AMS 5 (Thermo 55i, installed 2021); WBEA reports analyser replacements at AMS 5 and AMS 23 in September 2024. AMS 5 has a meteorological tower with sensors at 20–90 m; its 90 minus 20 m temperature difference is the stability measure.

Table 2
Observational datasets and temporal design of the methane-monitoring analysis.

Dataset	Variables (units)	Source	Resolution	Years
Methane	CH_4 (ppm)	WBEA continuous monitoring, final validated data	1 h	2021–2026
Surface wind	Speed ($m\,s^{-1}$) and direction (°)	WBEA continuous monitoring	1 h	2021–2026
Air temperature	Temperature (°C) at 2–90 m; AMS 5 tower gradient	WBEA continuous monitoring	1 h	2021–2026
Boundary-layer height	ERA5 boundary-layer height (m)	Copernicus Climate Change Service, ERA5	1 h	2021–2026
Station locations	Latitude, longitude (WGS84)	WBEA site documents (2024)	–	–

Temporal partition	Dates (MST, hour-ending)	Use
Development	1 Jan 2021 01:00 – 2 Dec 2025 00:00	Exploratory analysis and method development
Reference	1 Jan 2021 – 31 Dec 2023	Fixed station medians, thresholds and stability scaling
Confirmation	1 Jan 2026 01:00 – 1 Oct 2026 00:00	Independent confirmation (sealed until methods were frozen)

All timestamps are Mountain Standard Time (UTC−7, no daylight saving) and refer to the end of the hourly averaging period, as in WBEA reporting. Missing-value handling, station and pair eligibility, and comparison windows are described in Section 3.

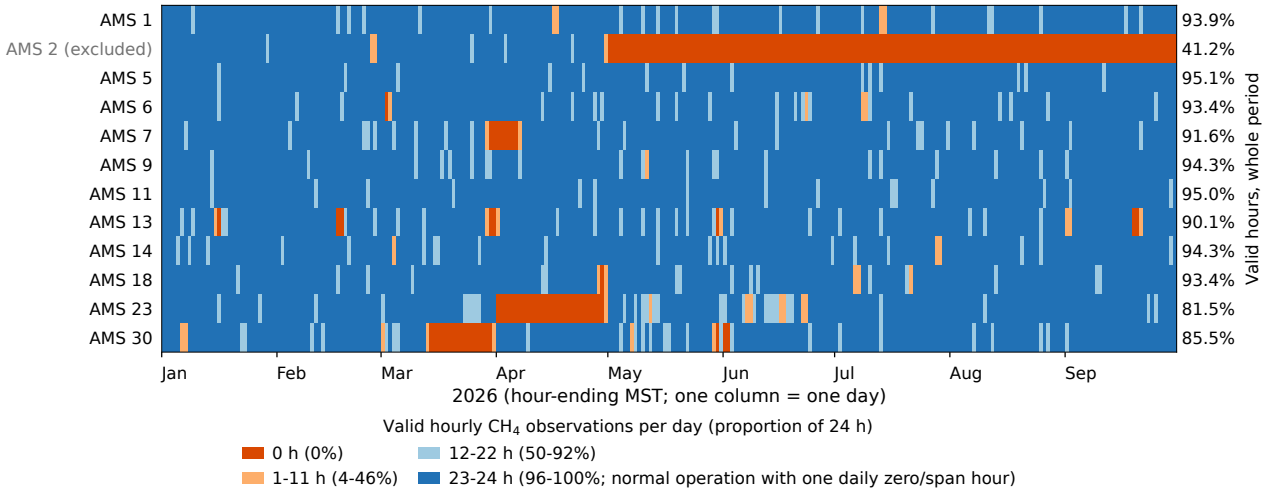


Figure 2: Temporal availability of hourly methane concentration observations at the WBEA monitoring stations from January to September 2026. The colour scale represents the proportion of valid hourly observations within each day. Missing measurements are distinguished from valid concentration measurements.